# Numerical Computation of the Ignorance Gap for the Radar Beam-Search Case Study

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**Abstract**

We numerically compute the ignorance-gap distribution $IG(x)$ for the finite-POMDP radar beam-search case study of `AdaptiveDesign.tex` §9, on a single-thread laptop, using exact enumeration and no Monte Carlo approximation. The computation produces closed-form nats values for (a) the oracle value $V_{\text{oracle}}(x)$ for every cell, (b) the fixed-design optimum $V_{\text{fixed}}$ with explicit optimal sensor sequence $s^\star$, (c) the per-cell fixed-design value $\Phi(x, s^\star, c)$, and (d) the per-cell ignorance gap $IG(x)$ in both the narrow-beam and wide-beam regimes. The document also clarifies one conceptual subtlety: the terminal reward used in the discrete-state numerical problem is the mutual information $I(z; y_{1:N})$, not a literal Fisher-information log-determinant, and the two agree only in the continuous-state Gaussian-posterior limit.

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# 1 Purpose and scope

The companion document `AdaptiveDesign.tex` develops the ignorance gap

$$IG(\boldsymbol{x}) = V_{\text{oracle}}(\boldsymbol{x}) - V_{\text{fixed}}(\boldsymbol{x}; s^\star)$$

as an EVPI upper bound on the value of any adaptive policy. Section 9 of that document analyzes the radar beam-search problem in two regimes—narrow beam and wide beam—and derives closed-form predictions for $\mathbb{E}[IG]$ in the idealized $(p_{\text{D}}, p_{\text{FA}}) \to (1, 0)$ limit.

This note replaces the idealized estimates with *exact finite-noise numerics* by enumerating all finite POMDP quantities directly. The problem size ($K = 16$ states, $M = 16$ actions, $|\mathcal{Y}| = 2$ observations, horizon $N = 4$) is small enough that no approximation, sampling, or solver beyond exhaustive enumeration is needed. The intent is to (i) verify the qualitative predictions, (ii) quantify the finite-noise correction, (iii) expose the full distribution of $IG(x)$, and (iv) surface one conceptual subtlety that had been glossed in the main document.

## 2 Discrete-state reformulation of the terminal reward

The main document defines the terminal reward as

$$\Phi(z, \theta_{1:N}, c) \;=\; \ln K - H(z \mid y_{1:N})$$

and notes that this equals $\ln\det \boldsymbol{J}_N$ "up to an additive constant under the categorical's softmax parametrization." That equivalence is loose in the discrete-state setting: there is no continuous parameter $x$ to differentiate, so Fisher information as a Hessian is ill-defined. The honest discrete analogue is the *mutual information*

$$\Phi(z, s, c) \;=\; I(z;\, y_{1:N} \mid s, c) \;=\; \ln K - \mathbb{E}_{y|z,s,c}[H(z \mid y_{1:N}, s, c)], \tag{1}$$

which is well-defined for any finite $z$-alphabet, matches the "information-gain" reading of $\Phi$, and coincides with $\ln\det \boldsymbol{J}_N$ only in the continuous-state Gaussian-posterior limit (where a softmax-logit Gaussian approximation of the posterior becomes exact).

For the numerical work we use $\Phi$ as defined in (1) throughout. No Hessian or log-determinant is ever computed; everything reduces to probabilities and entropies on a finite alphabet.

## 3 Problem parameters and beam model

**State.** $K = 16$ cells equally spaced on a ring: $\varphi(z) = 2\pi(z-1)/K$ for $z \in \{1, \dots, K\}$. Prior $p_0(z) = 1/K$.

**Actions.** $M = 16$ pointings: $\theta_a = \varphi(a)$ for $a \in \{1, \dots, K\}$. The action at step $k$ is an index $s_k \in \{1, \dots, K\}$ denoting the cell the beam is centered on.

**Observations.** Binary $y_k \in \{0, 1\}$ (detection / miss) with Bernoulli likelihood

$$p(y_k = 1 \mid z, s_k) = \begin{cases} p_{\mathrm{D}} & \text{if } z \text{ lies in the beam centered at cell } s_k, \\ p_{\mathrm{FA}} & \text{otherwise.} \end{cases} \tag{2}$$

**Beam model (top-hat).** The geometry $c$ is summarized by a single integer $m$ = number of cells covered by the main lobe. Let $d(a, z) = \min(\mathrm{mod}(z-a, K),\, \mathrm{mod}(a-z, K))$ be the angular distance (in cells) between pointing $a$ and cell $z$. Then

$$z \text{ is in beam centered at } a \iff d(a, z) \le \lfloor (m-1)/2 \rfloor. \tag{3}$$

Two regimes are reported:

- **Narrow-beam:** $m = 1$. Beam at cell $a$ covers only cell $a$.
- **Wide-beam:** $m = K/2 = 8$. The even-$m$ convention in our code yields a beam covering 7 cells (the target cell plus 3 on each side). This is an order-one rounding of $K/2$; qualitative behavior matches the wide-beam regime of the main document.

**Probabilities.** $p_{\mathrm{D}} = 0.9$, $p_{\mathrm{FA}} = 0.05$. These are finite; the $(p_{\mathrm{D}}, p_{\mathrm{FA}}) \to (1, 0)$ closed-form predictions of the main document are upper-bound limits.

## 4 Posterior and entropy formulas

Given a sensor sequence $s_{1:N}$ and observations $y_{1:N}$, the posterior on $z$ is

$$b(z \mid y_{1:N}, s_{1:N}) \;=\; \frac{p_0(z) \prod_{k=1}^{N} p(y_k \mid z, s_k)}{\sum_{z'} p_0(z') \prod_{k=1}^{N} p(y_k \mid z', s_k)}. \tag{4}$$

Its Shannon entropy (nats) is

$$H(z \mid y_{1:N}, s_{1:N}) = -\sum_{z=1}^{K} b(z \mid y_{1:N}, s_{1:N}) \ln b(z \mid y_{1:N}, s_{1:N}). \tag{5}$$

The state-conditional value of $\Phi$ is the observation-averaged information gain:

$$\Phi(x, s_{1:N}, c) = \sum_{y_{1:N} \in \{0,1\}^N} p(y_{1:N} \mid x, s_{1:N}) \left[\ln K - H(z \mid y_{1:N}, s_{1:N})\right] \tag{6}$$

with

$$p(y_{1:N} \mid x, s_{1:N}) = \prod_{k=1}^{N} p(y_k \mid x, s_k). \tag{7}$$

The sum in (6) is over $2^N = 16$ terms (a direct enumeration); the posterior in (4) sums over $K = 16$ terms. Per $(x, s_{1:N})$ evaluation: $16 \times (K \cdot N + K) \approx 16 \times 80 = 1280$ elementary operations.

# 5 Oracle value $V_{\text{oracle}}(x)$

## 5.1 Definition

$$V_{\text{oracle}}(x) = \max_{s_{1:N}} \Phi(x, s_{1:N}, c).$$

## 5.2 Algorithm

For each of the $K = 16$ values of $x$, enumerate all $K^N = 65{,}536$ sensor sequences and retain the maximum. Cost: $16 \times 65{,}536 \times 1280 \approx 1.3 \times 10^9$ elementary operations. Julia single-threaded runtime: a few seconds.

## 5.3 Symmetry prediction

The problem has a dihedral $D_K$ symmetry (ring cyclic rotations plus one reflection) under the uniform prior and reflection-symmetric beam. Consequently $V_{\text{oracle}}(x)$ is *exactly* constant across $x$: if $s^\star$ is oracle-optimal for $x = z_0$, then $\sigma \cdot s^\star$ is oracle-optimal for $\sigma \cdot z_0$ with the same value.

## 5.4 Numerical result

| $x$ | $V_{\text{oracle}}(x)$ | $s^\star(x)$ |
|---|---|---|
| 1, …, 16 | 2.579634 | narrow: $[x, x, x, x]$ |
| 1, …, 16 | 1.923462 | wide: (see text) |

Table 1: Numerical $V_{\text{oracle}}(x)$ in nats, for both regimes. Constant to 15 decimal places over $x$—the residual variation is machine $\epsilon$.

**Narrow regime.** Oracle strategy is trivially $s^\star(x) = (x, x, x, x)$: point at the known target every step. Per-step on-target Bernoulli is $p_{\text{D}} = 0.9$ (vs. $p_{\text{FA}} = 0.05$ for every other cell).

**Wide regime.** Oracle strategy is non-trivial: for $x = 1$, numerical $s^\star = [3, 14, 14, 4]$. Each of these pointings places cell 1 inside its 7-cell beam, so every measurement is on-target from cell 1's perspective; but the three distinct pointings exclude complementary non-1 cells. Specifically,

$$\begin{aligned} \text{beam at 3 excludes } &\{7, 8, 9, 10, 11, 12, 13\}, \\ \text{beam at 14 excludes } &\{2, 3, 4, 5, 6, 7, 8, 9, 10\}, \\ \text{beam at 4 excludes } &\{8, 9, 10, 11, 12, 13, 14, 15, 16\}. \end{aligned}$$

The union covers all cells except cell 1, so three *distinct* detections together triangulate the target. The oracle exploits the geometric overlap structure—a capability that narrow-beam oracle (single-cell beams) cannot use.

### 5.5 Why $V_{\text{oracle}}^{\text{narrow}} > V_{\text{oracle}}^{\text{wide}}$

Narrow beams separate a single cell from all others at $p_{\text{D}}/p_{\text{FA}} = 18{:}1$ per measurement. Wide beams separate $m = 7$ cells from the $K - m = 9$ remaining at the same per-shot ratio, but the oracle can no longer isolate a single cell with one measurement because seven cells share the "in-beam" label. Narrow per-step information is higher; oracle is stronger. Specifically $V_{\text{oracle}}^{\text{narrow}} = 2.580$ nats vs. $V_{\text{oracle}}^{\text{wide}} = 1.923$ nats, both bounded above by the prior entropy $\ln K = 2.773$ nats.

## 6 Fixed-design value $V_{\text{fixed}}$

### 6.1 Definition

$$V_{\text{fixed}} = \max_{s_{1:N}} \frac{1}{K} \sum_{x=1}^{K} \Phi(x, s_{1:N}, c), \qquad s^\star = \arg\max_{s_{1:N}} \frac{1}{K} \sum_{x=1}^{K} \Phi(x, s_{1:N}, c).$$

### 6.2 Algorithm

Enumerate all $K^N = 65{,}536$ sensor sequences; for each, average $\Phi$ over the uniform prior and track the maximum. Total per-sequence cost includes the inner $K = 16$ state loop: $65{,}536 \times 16 \times 1280 \approx 1.3 \times 10^9$ operations. Julia runtime: a few seconds.

### 6.3 Numerical result

| regime | $s^\star$ | $V_{\text{fixed}}$ (nats) |
|---|---|---|
| narrow | $[1, 13, 9, 7]$ (any 4 distinct cells, up to $D_K$ orbit) | 0.498683 |
| wide | $[1, 3, 13, 15]$ (4 spread pointings) | 1.410526 |

Table 2: Fixed-design optimum, exhaustive enumeration.

### 6.4 Per-cell breakdown of $\Phi(x, s^\star, c)$

Unlike $V_{\text{oracle}}(x)$, the fixed-design conditional value $\Phi(x, s^\star, c)$ is *not* constant over $x$ because a deterministic $s^\star$ breaks the ring symmetry. Table 3 shows the per-cell breakdown for both regimes.

| $x$ | $\Phi(x, s^\star)$ narrow | $\Phi(x, s^\star)$ wide |
|---|---|---|
| 1 | 0.976697 | 1.540807 |
| 2 | 0.339346 | 1.540807 |
| 3 | 0.339346 | 1.431264 |
| 4 | 0.339346 | 1.431264 |
| 5 | 0.339346 | 1.354544 |
| 6 | 0.339346 | 1.354544 |
| 7 | 0.976697 | 1.164713 |
| 8 | 0.339346 | 1.164713 |
| 9 | 0.976697 | 1.164713 |
| 10 | 0.339346 | 1.354544 |
| 11 | 0.339346 | 1.354544 |
| 12 | 0.339346 | 1.431264 |
| 13 | 0.976697 | 1.431264 |
| 14 | 0.339346 | 1.540807 |
| 15 | 0.339346 | 1.540807 |
| 16 | 0.339346 | 1.767821 |
| avg | 0.498683 | 1.410526 |

Table 3: Conditional fixed-design value $\Phi(x, s^\star, c)$ for each $x$. Narrow: 2 distinct values (probed / unprobed). Wide: 5 distinct values (the cell's position relative to $s^\star = [1, 3, 13, 15]$). Averages match $V_{\text{fixed}}$ exactly.

**Narrow regime interpretation.** With $s^\star = [1, 13, 9, 7]$, the four probed cells each see one on-target beam (which gives a detection with probability $p_{\text{D}} = 0.9$), leading to high information about whether $z$ equals that cell. The 12 unprobed cells see four off-target beams and the information content is much lower.

$$\Phi(\text{probed}) = 0.976697 \text{ nats}$$
$$\Phi(\text{unprobed}) = 0.339346 \text{ nats}$$

The prior-weighted average is $V_{\text{fixed}} = (4/16)(0.977) + (12/16)(0.339) = 0.499$ nats.

**Wide regime interpretation.** With $s^\star = [1, 3, 13, 15]$ (four spread pointings) and $m = 7$-cell beams, different cells see different numbers of on-target beams:

- Cell 16 lies in *all four* beams (closest to all four pointings by ring wraparound) and receives the highest $\Phi = 1.768$.
- Cells 1, 2, 14, 15 lie in three beams and have $\Phi = 1.541$.
- Cells 3, 4, 12, 13 lie in two beams and have $\Phi = 1.431$.
- Cells 5, 6, 10, 11 lie in one beam and have $\Phi = 1.355$.
- Cells 7, 8, 9 (the "back" of the ring, between pointings 3 and 13) lie in *zero* beams: $\Phi = 1.165$.

The five-level structure directly reflects the in-beam count, independent of cell identity.

# 7 Ignorance gap $IG(x)$

## 7.1 Definition and computation

$$IG(x) = V_{\text{oracle}}(x) - \Phi(x, s^\star, c).$$

Since $V_{\text{oracle}}(x)$ is constant over $x$ (exact symmetry) and $\Phi(x, s^\star, c)$ has been tabulated per $x$, this is a direct subtraction.

### 7.2 Narrow regime

| $x$ | $V_{\text{oracle}}$ | $\Phi(x, s^\star)$ | $IG(x)$ |
|---|---|---|---|
| 1 | 2.579634 | 0.976697 | 1.602937 |
| 2 | 2.579634 | 0.339346 | 2.240288 |
| 3 | 2.579634 | 0.339346 | 2.240288 |
| 4 | 2.579634 | 0.339346 | 2.240288 |
| 5 | 2.579634 | 0.339346 | 2.240288 |
| 6 | 2.579634 | 0.339346 | 2.240288 |
| 7 | 2.579634 | 0.976697 | 1.602937 |
| 8 | 2.579634 | 0.339346 | 2.240288 |
| 9 | 2.579634 | 0.976697 | 1.602937 |
| 10 | 2.579634 | 0.339346 | 2.240288 |
| 11 | 2.579634 | 0.339346 | 2.240288 |
| 12 | 2.579634 | 0.339346 | 2.240288 |
| 13 | 2.579634 | 0.976697 | 1.602937 |
| 14 | 2.579634 | 0.339346 | 2.240288 |
| 15 | 2.579634 | 0.339346 | 2.240288 |
| 16 | 2.579634 | 0.339346 | 2.240288 |
| mean | | | $\mathbb{E}[IG] = 2.080950$ |
| std | | | 0.285032 |
| distinct values | | | 2 |

Table 4: Narrow-regime $IG(x)$ table. The 4 probed cells (1, 7, 9, 13) have $IG = 1.603$; the 12 unprobed cells have $IG = 2.240$.

The two-value distribution in the narrow regime is a direct consequence of the 4-vs-12 partition created by $s^\star$. Cells inside $s^\star$ have lower IG because the fixed design does give them one on-target probe, reducing the gap to the oracle. Cells outside $s^\star$ are treated identically (all-off-target beams) and hence have identical IG.

### 7.3 Wide regime

Across the wide regime the gap varies by a factor of nearly 5× (from 0.156 to 0.759), with cell 16 the most-covered and cells 7–9 the "shadow" of the spread pointings $[1, 3, 13, 15]$.

## 8 Why $IG(x)$ is not constant across $x$

At first glance the non-constancy is surprising: the prior is uniform, the ring is $D_K$-symmetric, the beam pattern is rotation-equivariant—so shouldn't every cell look equivalent?

The resolution is that the oracle and the fixed design treat the symmetry group differently.

**Oracle.** By definition, $V_{\text{oracle}}(x) = \max_s \Phi(x, s, c)$. For each $x$, the oracle *re-optimizes* and picks the best sensor sequence for that particular $x$. Under the group action $\sigma \in D_K$, the optimal sensor sequence for $\sigma \cdot x$ is $\sigma \cdot s^\star(x)$, and $\Phi(\sigma \cdot x, \sigma \cdot s^\star(x), \sigma \cdot c) = \Phi(x, s^\star(x), c)$ by equivariance. Hence $V_{\text{oracle}}(x)$ is truly constant over $x$.

**Fixed design.** The best fixed sensor sequence $s^\star_{\text{fixed}}$ is chosen once and used for every $x$. For a single deterministic $s^\star$, the value $\Phi(x, s^\star, c)$ under the group action becomes $\Phi(\sigma \cdot x, s^\star, c)$, which does *not* equal $\Phi(x, s^\star, c)$ unless $s^\star$ is $\sigma$-invariant. No single $s^\star$ is invariant under the full

| $x$ | $V_{\text{oracle}}$ | $\Phi(x, s^\star)$ | $IG(x)$ |
|---|---|---|---|
| 1 | 1.923462 | 1.540807 | 0.382655 |
| 2 | 1.923462 | 1.540807 | 0.382655 |
| 3 | 1.923462 | 1.431264 | 0.492197 |
| 4 | 1.923462 | 1.431264 | 0.492197 |
| 5 | 1.923462 | 1.354544 | 0.568918 |
| 6 | 1.923462 | 1.354544 | 0.568918 |
| 7 | 1.923462 | 1.164713 | 0.758748 |
| 8 | 1.923462 | 1.164713 | 0.758748 |
| 9 | 1.923462 | 1.164713 | 0.758748 |
| 10 | 1.923462 | 1.354544 | 0.568918 |
| 11 | 1.923462 | 1.354544 | 0.568918 |
| 12 | 1.923462 | 1.431264 | 0.492197 |
| 13 | 1.923462 | 1.431264 | 0.492197 |
| 14 | 1.923462 | 1.540807 | 0.382655 |
| 15 | 1.923462 | 1.540807 | 0.382655 |
| 16 | 1.923462 | 1.767821 | 0.155641 |
| mean | | | $\mathbb{E}[IG] = 0.512935$ |
| std | | | 0.161468 |
| distinct values | | | 5 |

Table 5: Wide-regime $IG(x)$ table. Five distinct values, indexed by how many of the four $s^\star$ beams cover cell $x$.

$D_K$ orbit, so $\Phi(x, s^\star, c)$ must vary across the orbit—and by size of its stabilizer, it takes on a number of distinct values equal to $K$ divided by the orbit length of $s^\star$.

**Consequence.** The deterministic fixed-design value decomposes the $K$ cells according to their position relative to $s^\star$:

- Narrow regime: 4 cells in $s^\star$, 12 outside. Two-valued $IG$.
- Wide regime: 5 classes depending on how many $s^\star$-beams cover each cell (0, 1, 2, 3, or 4 beams). Five-valued $IG$.

The prior-expected IG is

$$\mathbb{E}[IG] = V_{\text{oracle}} - V_{\text{fixed}} = V_{\text{oracle}} - \mathbb{E}_x[\Phi(x, s^\star, c)],$$

which is a single scalar, but the orbit-dependent distribution of $\Phi(x, s^\star, c)$ around that mean gives the per-cell spread of $IG(x)$.

**Orbit averaging would restore constancy.** A *randomized* fixed design that applies $s^\star$ with uniform probability over its $D_K$ orbit would give $\mathbb{E}_\sigma[\Phi(x, \sigma \cdot s^\star, c)]$ constant in $x$ by symmetry. The deterministic design sacrifices this for the best single-realization value. In a sense, the variability of $IG(x)$ is the price of deterministic commitment.

# 9 Comparison with the closed-form predictions

## 9.1 Ideal-limit predictions

The main document's §9 derives, under $(p_{\mathrm{D}}, p_{\mathrm{FA}}) \to (1, 0)$:

$$V_{\mathrm{oracle}} \approx \ln K = 2.773 \text{ nats}, \tag{8}$$

$$V_{\mathrm{fixed}}^{\mathrm{narrow}} \approx \ln K - \frac{K-N}{K} \ln(K-N) = 0.909 \text{ nats}, \tag{9}$$

$$V_{\mathrm{fixed}}^{\mathrm{wide}} \approx \ln(2N) = 2.079 \text{ nats}, \tag{10}$$

$$\mathbb{E}[IG]^{\mathrm{narrow}} \approx \frac{K-N}{K} \ln(K-N) = 1.864 \text{ nats}, \tag{11}$$

$$\mathbb{E}[IG]^{\mathrm{wide}} \approx \ln\big(K/(2N)\big) = 0.693 \text{ nats}. \tag{12}$$

## 9.2 Ideal vs. finite-noise

| quantity | ideal $(p_{\mathrm{D}} \to 1, p_{\mathrm{FA}} \to 0)$ | finite $p_{\mathrm{D}} = 0.9, p_{\mathrm{FA}} = 0.05$ | ratio |
|---|---|---|---|
| $V_{\mathrm{oracle}}^{\mathrm{narrow}}$ | 2.773 | 2.580 | 0.93 |
| $V_{\mathrm{fixed}}^{\mathrm{narrow}}$ | 0.909 | 0.499 | 0.55 |
| $\mathbb{E}[IG]^{\mathrm{narrow}}$ | 1.864 | 2.081 | 1.12 |
| $V_{\mathrm{oracle}}^{\mathrm{wide}}$ | 2.773 | 1.923 | 0.69 |
| $V_{\mathrm{fixed}}^{\mathrm{wide}}$ | 2.079 | 1.411 | 0.68 |
| $\mathbb{E}[IG]^{\mathrm{wide}}$ | 0.693 | 0.513 | 0.74 |

Table 6: Idealized closed-form predictions vs. exact finite-noise numerics.

## 9.3 Interpretation

- *Narrow regime.* $\mathbb{E}[IG]$ is *larger* at finite noise than predicted (2.081 vs. 1.864). The reason: both $V_{\mathrm{oracle}}$ and $V_{\mathrm{fixed}}$ shrink as noise increases, but $V_{\mathrm{fixed}}$ shrinks more aggressively (factor 0.55) than $V_{\mathrm{oracle}}$ (factor 0.93). The oracle can always point at the target and extract information efficiently even at finite noise; the fixed design wastes most of its budget on cells that aren't the target, and each off-target beam gives a weak Bernoulli($p_{\mathrm{FA}}$) whose information content degrades with increasing $p_{\mathrm{FA}}$. The gap *widens* as noise grows.

- *Wide regime.* All three quantities shrink by roughly the same factor ($\approx 0.7$). The gap is dominated by the coverage structure, which degrades oracle and fixed-design roughly symmetrically as noise increases.

The qualitative structure predicted by the main document—$\mathbb{E}[IG]^{\mathrm{narrow}} > \mathbb{E}[IG]^{\mathrm{wide}}$—holds unambiguously: $2.08 > 0.51$. The numerical margin is even larger than the ideal-limit prediction.

# 10 Summary of IG statistics

The coefficient of variation std/$\mathbb{E}[IG]$ is larger for the wide regime (0.31 vs. 0.14), consistent with the richer 5-level structure: even though $\mathbb{E}[IG]$ is smaller, its distribution across cells is *more* heterogeneous in relative terms. If one were to deploy an adaptive policy, the cells with the highest $IG(x)$ (indices 7, 8, 9 in the wide regime) would be the ones where adaptation could in principle help the most.

| statistic | narrow | wide |
|---|---|---|
| $\mathbb{E}[IG]$ | 2.080950 | 0.512935 |
| $\mathrm{std}[IG]$ | 0.285032 | 0.161468 |
| $\min[IG]$ | 1.602937 | 0.155641 |
| $\max[IG]$ | 2.240288 | 0.758748 |
| max / min | 1.40 | 4.87 |
| distinct values | 2 | 5 |
| CV = std/$\mathbb{E}$ | 0.14 | 0.31 |

Table 7: Distributional statistics of $IG(x)$.

# 11 Computational cost summary

All reported quantities come from exact enumeration, no sampling. Per-regime:

- $V_{\text{oracle}}$: $K \cdot K^N \cdot 2^N \cdot (KN + K)$ elementary operations $= 16 \cdot 65{,}536 \cdot 16 \cdot 80 \approx 1.3 \times 10^9$.
- $V_{\text{fixed}}$: identical count.

Total for both regimes: $\approx 5 \times 10^9$ elementary operations. Julia 1.12 single-threaded completes both in well under a minute on a standard laptop. RAM footprint: under 20 MB (the largest array is one posterior vector of length $K = 16$ per outcome, recomputed on the fly).

# 12 Conceptual reading of the results

## 12.1 The oracle is symmetric; the deterministic fixed design is not

The equality $V_{\text{oracle}}(x) = \text{const}$ is an exact symmetry fact. The non-constancy of $IG(x)$ comes entirely from $\Phi(x, s^\star, c)$, the per-$x$ fixed-design value, which depends on the $x$-orbit position relative to a single chosen $s^\star$. The narrow regime has a simple binary decomposition (probed vs. unprobed); the wide regime has a richer 5-level structure determined by cell-to-beam overlap count. Both are fully predicted by the orbit structure.

## 12.2 Finite noise widens the narrow-regime gap

The ideal-limit prediction $\mathbb{E}[IG]^{\text{narrow}} \approx 1.86$ nats becomes 2.08 nats at finite noise because $V_{\text{fixed}}$ shrinks faster than $V_{\text{oracle}}$. This is a second-order effect not visible in the idealized analysis, but structurally expected: the fixed design's off-target beams have $p_{\text{FA}}$-dominated likelihoods that degrade sharply as $p_{\text{FA}}$ grows from 0.

## 12.3 $\mathbb{E}[IG]$ as a diagnostic, not a promise

In the narrow regime, $\mathbb{E}[IG] = 2.08$ nats is substantial (EVPI upper bound of $\sim 2.5\times$ uncertainty-volume reduction), but as the main document argues, adaptation cannot realize any of this in a narrow-beam setting: with single-cell beams, adaptive and fixed designs both visit $N$ cells in $N$ steps, in the same expected order under the uniform prior. Section 9 of `AdaptiveDesign.tex` predicts $V^{\text{narrow}}_{\text{adaptive}} = V^{\text{narrow}}_{\text{fixed}} = 0.499$ nats. Verifying this—and verifying the tight-EVPI result $V^{\text{wide}}_{\text{adaptive}} - V^{\text{wide}}_{\text{fixed}} = \mathbb{E}[IG]^{\text{wide}} = 0.51$ nats—requires the full finite-horizon Bellman DP, which is the next computational step.

# 13 Exact Bellman DP and $V_{\text{adaptive}}$

## 13.1 Algorithm

The terminal reward $\Phi = \ln K - H(z \mid y_{1:N})$ is a nonlinear functional of the terminal belief, so Sondik's piecewise-linear-convex representation does not apply directly. We instead do backward induction on the belief tree reachable from the uniform prior. The recursion is

$$W_k(b) = \max_{a \in \{1,\ldots,K\}} \sum_{y \in \{0,1\}} P(y \mid b, a) \cdot W_{k+1}\big(T(b, a, y)\big), \qquad W_{N+1}(b) = \ln K - H(b), \tag{13}$$

where $T(b, a, y)$ is the Bayesian update of $b$ under observation $y$ when taking action $a$.

At depth $d$ from root the recursion branches by ($K$ actions) $\times$ (2 outcomes), giving $(2K)^N = 32^4 \approx 10^6$ terminal evaluations in the worst case. Each evaluation is $O(K)$ for the posterior update and entropy. Total work: $\approx 10^7$ elementary operations per regime. Julia runtime: $\sim 0.1$ seconds. Memory peak: a handful of belief vectors of length $K$, so $< 1$ MB.

## 13.2 Numerical results

| | narrow ($m$=1) | wide ($m$=$K/2$=8) |
|---|---|---|
| $V_{\text{oracle}}$ | 2.579634 | 1.923462 |
| $V_{\text{adaptive}}$ | 0.619198 | 1.699767 |
| $V_{\text{fixed}}$ | 0.498683 | 1.410526 |
| $\mathbb{E}[IG]$ (EVPI bound) | 2.080950 | 0.512935 |
| $V_{\text{adaptive}} - V_{\text{fixed}}$ (realized gain) | 0.120515 | 0.289241 |
| EVPI saturation % | 5.8% | 56.4% |

Table 8: Exact Bellman-DP results with $K = 16$, $N = 4$, $p_{\text{D}} = 0.9$, $p_{\text{FA}} = 0.05$. All numbers in nats unless noted. "EVPI saturation" is the fraction $(V_{\text{adaptive}} - V_{\text{fixed}})/\mathbb{E}[IG]$, showing how much of the ignorance-gap upper bound the optimal adaptive policy actually realizes.

## 13.3 Optimal policy tree (first two levels)

**Narrow regime.** The depth-2 head of the DP-optimal policy tree:

| history | action |
|---|---|
| $\emptyset$ (root) | cell 1 |
| $y_1 = 0$ (miss) | cell 1 |
| $y_1 = 1$ (detect) | cell 16 |
| $(y_1, y_2) = (0, 0)$ | cell 1 |
| $(y_1, y_2) = (0, 1)$ | cell 16 |
| $(y_1, y_2) = (1, 0)$ | cell 16 |
| $(y_1, y_2) = (1, 1)$ | cell 16 |

The policy probes cell 1 at the first step and, if the observation was a miss, probes it *again* at step 2 (because at finite noise a single miss does not definitively rule out cell 1: the posterior on cell 1 only dropped from $1/16 = 0.0625$ to $\approx 0.007$). After a detection it switches to a different cell.

**Wide regime.** The depth-2 head:

| history | action |
|---|---|
| $\emptyset$ (root) | cell 1 |
| $y_1 = 0$ (miss) | cell 4 |
| $y_1 = 1$ (detect) | cell 13 |
| $(0, 0)$ | cell 4 |
| $(0, 1)$ | cell 3 |
| $(1, 0)$ | cell 3 |
| $(1, 1)$ | cell 4 |

The wide-beam adaptive policy shows visible binary-search structure: a detection at cell 1 (which covers cells $\{14, 15, 16, 1, 2, 3, 4\}$) narrows the support to roughly the top half of the ring; a miss narrows to the bottom half. Step 2 at cell 4 (which covers $\{1, \ldots, 7\}$) vs. cell 13 (which covers $\{10, \ldots, 16\}$) is consistent with bisecting the post-step-1 support.

### 13.4 Departures from idealized predictions

Both regimes diverge from the idealized predictions of `AdaptiveDesign.tex` §9 in the direction expected once finite Bernoulli noise is taken into account:

**Narrow regime predicted zero adaptive benefit; actually 0.12 nats.** The main document's §9 claim that "under the strictly narrow-beam restriction, $V_{\text{adaptive}} = V_{\text{fixed}}$" is exact only in the ideal $(p_{\text{D}}, p_{\text{FA}}) \to (1, 0)$ limit. At finite noise the adaptive policy does realize a small but strictly positive gain—5.8% of the EVPI ceiling. The mechanism is visible in the policy tree: after a detection the adaptive policy *switches* to a different cell (since the on-target cell is now likely resolved), whereas a pre-committed fixed schedule would continue visiting cells determined in advance. The benefit is modest because the fixed design with 4 distinct-cell pointings is *already* almost-optimally-chosen under the uniform prior, leaving adaptation little room.

**Wide regime predicted $\ln 2 \approx 0.69$ nats gain; actually 0.29 nats.** The §9 idealized prediction $V_{\text{adaptive}} - V_{\text{fixed}} = \ln(K/(2N)) = \ln 2$ is an upper bound assumed to be saturated when adaptive binary search can fully resolve. At finite $(p_{\text{D}}, p_{\text{FA}})$ full resolution isn't achievable in $N = 4$ noisy binary measurements (the information bound $4 \ln 2 = 2.77$ nats is not reached by either side). Adaptive achieves 1.70 nats, fixed 1.41, leaving a realized gap of 0.29 nats—56% of the EVPI. The direction matches prediction but the magnitude is reduced by noise.

**EVPI tightness structure is actually reversed from the idealization.** The idealized analysis suggested the wide regime had tight EVPI and the narrow regime slack EVPI. Numerically this is qualitatively preserved (56% vs. 6% saturation), but the narrow regime is *more* slack than predicted (expected 0%, got 6%, which is in the right direction), and the wide regime is *less* tight than predicted (expected 100%, got 56%). The quantitative message is that *EVPI is a genuinely loose upper bound at finite noise in all regimes we checked*, and the realized adaptive benefit is always strictly less than $\mathbb{E}[IG]$. This strengthens rather than weakens the case for running an exact DP when adaptation is being considered.

### 13.5 Key quantitative takeaway

At the joint $(\boldsymbol{c}, \pi)$ optimum (wide beam, adaptive policy), finite-noise numerics give

$$V^{\star}_{\text{adaptive}} - V^{\star}_{\text{fixed}} = 0.289 \text{ nats.} \tag{14}$$

This is the *actual* dividend of adopting the full adaptive framework at this problem size—measurable, positive, but about half of what the idealized closed-form suggested. The ignorance-gap go/no-go gate ($\mathbb{E}[IG] > 1$ nat in the narrow regime) would have recommended proceeding with the DP; post-solve, we see the realized gain was much smaller but still present.

# 14 Joint geometry–policy optimization (Gaussian beam)

The top-hat beam of §3 is discrete in $m$, so gradient-based outer optimization of the geometry is not directly meaningful. We now replace the beam model with a continuously parametrized Gaussian and sweep $w$ over a grid to map the joint $(\boldsymbol{c}, \pi)$ landscape.

## 14.1 Gaussian-beam likelihood

Parametrize the geometry by a single continuous scalar $w > 0$ (beamwidth):

$$G_w(\alpha) = \exp\bigl(-\alpha^2/(2w^2)\bigr), \qquad p(y_k = 1 \mid z, a; w) = p_{\mathrm{FA}} + (p_{\mathrm{D}} - p_{\mathrm{FA}})\, G_w\bigl(\theta_a - \varphi(z)\bigr), \tag{15}$$

with $\alpha$ wrapped to $[-\pi, \pi]$. All downstream quantities—$\Phi(x, s, w)$, $V_{\mathrm{oracle}}(w)$, $V_{\mathrm{fixed}}(w)$, $V_{\mathrm{adaptive}}(w)$—are smooth in $w$. We sweep $w$ on a grid and at each value compute all three values exactly by the same algorithms as §5–7 (symmetry-reduced enumeration for oracle/fixed, backward induction for the DP).

## 14.2 Sweep results

The total sweep (19 grid points, $w \in [0.05, \pi]$) completed in 122 seconds on a single Julia thread.

| $w$ (rad) | $V_{\mathrm{oracle}}$ | $V_{\mathrm{adaptive}}$ | $V_{\mathrm{fixed}}$ | $\mathbb{E}[IG]$ | realized gain | sat. % |
|---|---|---|---|---|---|---|
| 0.05 | 2.5796 | 0.6192 | 0.4987 | 2.0810 | 0.1205 | 5.8 |
| 0.10 | 2.5795 | 0.6191 | 0.4986 | 2.0809 | 0.1206 | 5.8 |
| 0.15 | 2.5666 | 0.6176 | 0.4924 | 2.0742 | 0.1252 | 6.0 |
| 0.20 | 2.4792 | 0.6411 | 0.4931 | 1.9861 | 0.1480 | 7.5 |
| 0.25 | 2.3035 | 0.6875 | 0.5346 | 1.7689 | 0.1529 | 8.6 |
| 0.30 | 2.1210 | 0.7335 | 0.5889 | 1.5320 | 0.1445 | 9.4 |
| 0.35 | 1.9500 | 0.7704 | 0.6358 | 1.3142 | 0.1347 | 10.2 |
| 0.40 | 1.8023 | 0.8059 | 0.6729 | 1.1294 | 0.1330 | 11.8 |
| 0.50 | 1.5714 | 0.8522 | 0.7301 | 0.8414 | 0.1222 | 14.5 |
| **0.60** | **1.3922** | **0.8647** | 0.7571 | 0.6351 | 0.1077 | 17.0 |
| 0.70 | 1.2426 | 0.8532 | **0.7576** | 0.4850 | 0.0956 | 19.7 |
| 0.80 | 1.1192 | 0.8264 | 0.7480 | 0.3712 | 0.0784 | 21.1 |
| 1.00 | 0.9118 | 0.7470 | 0.6968 | 0.2150 | 0.0502 | 23.4 |
| 1.25 | 0.7041 | 0.6305 | 0.6016 | 0.1025 | 0.0290 | 28.3 |
| 1.50 | 0.5819 | 0.5159 | 0.4983 | 0.0836 | 0.0176 | 21.1 |
| 1.75 | 0.5284 | 0.4146 | 0.4034 | 0.1250 | 0.0112 | 9.0 |
| 2.00 | 0.4698 | 0.3310 | 0.3234 | 0.1464 | 0.0076 | 5.2 |
| 2.50 | 0.3415 | 0.2119 | 0.2079 | 0.1336 | 0.0041 | 3.0 |
| $\pi$ | 0.2117 | 0.1237 | 0.1215 | 0.0902 | 0.0022 | 2.4 |

Table 9: Full sweep of the Gaussian-beam geometry parameter. Joint optima in bold: $V^\star_{\mathrm{adaptive}} = 0.865$ nats at $w = 0.60$, $V^\star_{\mathrm{fixed}} = 0.758$ nats at $w = 0.70$. The rightmost column is the EVPI saturation fraction.

### 14.3 Observations

**Optimum beam width.** $V_{\text{adaptive}}$ peaks at $w^\star \approx 0.60$ rad. In HPBW units (FWHM of the Gaussian, $= 2w\sqrt{2\ln 2} \approx 2.355\,w$), this corresponds to HPBW$^\star \approx 1.41$ rad $\approx 3.6$ cells. This is neither the narrow-beam limit (HPBW $\ll 2\pi/K$) nor the wide-beam limit (HPBW $\sim \pi$). The joint optimum lies in an intermediate regime where each measurement resolves a small neighborhood of cells well enough that four measurements can collectively localize the target.

**Why $\mathbf{V_{oracle}}$ decreases monotonically in $w$.** The oracle knows $x$ and points the beam directly at it. Narrower beams give larger $p(y = 1 \mid z = x) - p(y = 1 \mid z = x')$ gap, hence more information per measurement. No finite-noise ceiling binds: as $w \to 0$, $V_{\text{oracle}} \to V_{\text{oracle}}^{\text{narrow}} = 2.58$ nats (recovering the Section 5 value for the $m = 1$ top-hat).

**Why $\mathbf{V_{adaptive}}$ and $\mathbf{V_{fixed}}$ have interior maxima.** At very narrow $w$, per-measurement information is high but *only for cells near the pointing.* With $N = 4$ pointings across $K = 16$ cells, fixed and adaptive both miss most of the prior mass most of the time. At very wide $w$, per-measurement information degrades and the best binary partition is weakly diagnostic. The interior optimum $w^\star \approx 0.6$ balances the two.

**Where the EVPI saturation is highest.** Saturation peaks at $w \approx 1.25$ rad (28.3%), far from the $\mathrm{V}_{\text{adaptive}}$ optimum. At large $w$ the EVPI itself is small (around 0.1 nats) and adaptation captures a relatively large fraction of a small gap. At the $\mathrm{V}_{\text{adaptive}}$ optimum ($w = 0.6$), saturation is only 17%—adaptation captures a small fraction of a moderately-sized EVPI, but the *absolute* realized gain at $w = 0.6$ (0.108 nats) is closer to the maximum of the realized-gain curve (0.153 at $w = 0.25$) than the saturation percentage alone would suggest.

**Joint optima for adaptive vs. fixed differ slightly.** $V_{\text{fixed}}$ peaks at $w = 0.70$, $V_{\text{adaptive}}$ at $w = 0.60$. The adaptive optimum prefers a slightly narrower beam because it can exploit the higher per-measurement information by routing each measurement based on earlier outcomes; the fixed design, committed to a non-adaptive schedule, benefits from a slightly wider beam that is more forgiving of pointing choices made in advance. The difference is small ($\Delta w = 0.1$, or about 14% relative) but consistent.

### 14.4 Envelope-theorem sanity check at the optimum

Around $w^\star = 0.6$, nearby values give

$$V_{\text{adaptive}}(0.5) = 0.8522, \quad V_{\text{adaptive}}(0.6) = 0.8647, \quad V_{\text{adaptive}}(0.7) = 0.8532.$$

Central finite difference at $w = 0.6$:

$$\frac{V_{\text{adaptive}}(0.7) - V_{\text{adaptive}}(0.5)}{2 \cdot 0.1} = \frac{0.8532 - 0.8522}{0.2} = 0.005 \text{ nats/rad}.$$

Essentially zero, consistent with $w = 0.6$ being a (coarse-grid) interior maximum. A finer grid or an analytic envelope-theorem gradient evaluation would refine this.

### 14.5 Representative optimal sensor sequences at $w^\star$

The symmetry-reduced fixed-design search at $w = 0.6$ returns

$$s^\star_{\text{fixed}} = [1, 5, 9, 13],$$

four pointings equally spaced around the ring (quarter-rotations of each other). Any $D_K$-orbit representative is equivalent. This differs from the top-hat narrow-beam result $s^\star = [1, 7, 9, 13]$

in that the Gaussian-beam likelihood favors maximally-separated pointings (no two within a beamwidth of each other), whereas the top-hat was indifferent to separation as long as all four cells were distinct.

### 14.6 Takeaway: EVPI diagnoses, joint DP decides

The sweep is the cleanest possible demonstration of *why joint geometry-policy optimization is irreducible.* The four columns of Table 9—$V_{\text{oracle}}$, $V_{\text{adaptive}}$, $V_{\text{fixed}}$, and $\mathbb{E}[IG]$—behave qualitatively differently as $w$ sweeps from 0 to $\pi$:

- $V_{\text{oracle}}$ is monotonically decreasing. The oracle always benefits from a narrower beam because it knows where to point.
- $V_{\text{adaptive}}$ has an interior maximum at $w = 0.60$ rad, value 0.865 nats.
- $V_{\text{fixed}}$ has an interior maximum at $w = 0.70$ rad, value 0.758 nats. Different from both the above.
- $\mathbb{E}[IG] = V_{\text{oracle}} - V_{\text{fixed}}$ inherits the monotone decrease of $V_{\text{oracle}}$ minus a shallow $V_{\text{fixed}}$ curve: $\mathbb{E}[IG]$ is monotonically decreasing (on this grid).
- The *realized* adaptive advantage $V_{\text{adaptive}} - V_{\text{fixed}}$ has yet another maximum, at $w = 0.25$ rad (0.153 nats).

Consequently:

**Necessary-but-not-sufficient claim:** $\mathbb{E}[IG](\boldsymbol{c})$ is an upper bound on $V_{\text{adaptive}}(\boldsymbol{c}) - V_{\text{fixed}}(\boldsymbol{c})$, but is typically a *loose* bound, and choosing $\boldsymbol{c}$ to maximize $\mathbb{E}[IG]$ does *not* maximize either $V_{\text{adaptive}}$ or the realized adaptive advantage. Restricting attention to a single beam family (here, the Gaussian family) and comparing the two extremes against the interior optimum makes this stark:

| $\boldsymbol{c}$ | $\mathbb{E}[IG]$ | $V_{\text{adaptive}}$ | $V_{\text{fixed}}$ | realized | sat. |
|---|---|---|---|---|---|
| Gaussian $w = 0.05$ (small-$w$ extreme) | **2.081** | 0.619 | 0.499 | **0.121** | 5.8% |
| Gaussian $w = \pi$ (large-$w$ extreme) | 0.090 | 0.124 | 0.122 | 0.002 | 2.4% |
| Gaussian $w = 0.60$ (in-family optimum) | 0.635 | **0.865** | **0.758** | 0.108 | **17.0**% |

Note the *non-monotone* relationship between $\mathbb{E}[IG]$ and the other columns. The small-$w$ extreme has the largest $\mathbb{E}[IG]$ (by a factor of $\sim 3$ over the in-family optimum and $\sim 23$ over the large-$w$ extreme), yet only the second-largest $V_{\text{adaptive}}$ and an EVPI saturation of only 5.8%. The large-$w$ extreme has both the smallest $\mathbb{E}[IG]$ and the smallest $V_{\text{adaptive}}$. The interior $w = 0.60$ *dominates* both extremes on $V_{\text{adaptive}}$ (the quantity we actually want to maximize) and on $V_{\text{fixed}}$, despite having neither the largest $\mathbb{E}[IG]$ nor the largest realized gain. Saturation alone is also misleading: $w = 0.60$ has the highest saturation, but $w = 0.05$ has higher realized gain ($0.121 > 0.108$) by virtue of its much larger EVPI ceiling.

Conclusion: no single column—$\mathbb{E}[IG]$, realized gain, or saturation—picks out the in-family optimum. Only $V_{\text{adaptive}}$ does, and only by direct evaluation. EVPI is a go/no-go gate; the joint DP, swept over an admissible family of $\boldsymbol{c}$, picks the optimum.

Use EVPI as the go/no-go gate. Use the joint DP, over the full admissible family of $\boldsymbol{c}$, to find the actual $(\boldsymbol{c}^\star, \pi^\star)$.

**Cross-family note.** Restricting to one family is a fair within-family test of the joint-DP claim. Across families, the comparison is also informative but not what is being claimed here: the top-hat family achieves higher $V_{\text{adaptive}}$ than any Gaussian (see the within-top-hat-family sweep in Section 14.7 below), because its sharper in-beam vs. out-of-beam likelihood contrast gives it

strictly higher per-measurement information. Which family is physically realizable (Gaussian, top-hat, Dolph–Chebyshev, phased-array-synthesized, etc.) is itself part of the geometry design.

### 14.7 Within-top-hat-family sweep

For completeness, we repeat the joint sweep within the *top-hat* beam family, parametrized by the integer main-lobe width $m \in \{1, 2, \ldots, K\}$ (cells per lobe). The coverage is a contiguous arc of $m$ cells centered on the pointed cell; per-cell likelihood is $p_{\mathrm{D}}$ inside the lobe and $p_{\mathrm{FA}}$ outside. Runtime is $\sim$23 s for the full $m \in \{1, \ldots, 16\}$ sweep on the reference laptop (same symmetry reductions as the Gaussian sweep).

Table 10: Within-top-hat-family joint sweep over integer $m$. In-family optimum for $V_{\mathrm{adaptive}}$ at $m^\star = 9$; in-family optimum for $V_{\mathrm{fixed}}$ at $m = 7$. Odd/even adjacent $m$ values produce identical coverage for the symmetric placements and hence identical numbers.

| $m$ | $V_{\mathrm{oracle}}$ | $V_{\mathrm{adaptive}}$ | $V_{\mathrm{fixed}}$ | $\mathbb{E}[IG]$ | realized | sat. % |
|---|---|---|---|---|---|---|
| 1 | 2.580 | 0.619 | 0.499 | **2.081** | 0.121 | 5.8% |
| 2 | 2.580 | 0.619 | 0.499 | 2.081 | 0.121 | 5.8% |
| 3 | 2.293 | 1.401 | 1.103 | 1.190 | 0.299 | 25.1% |
| 4 | 2.293 | 1.401 | 1.103 | 1.190 | 0.299 | 25.1% |
| 5 | 2.081 | 1.631 | 1.343 | 0.737 | 0.287 | 38.9% |
| 6 | 2.081 | 1.631 | 1.343 | 0.737 | 0.287 | 38.9% |
| 7 | 1.924 | 1.700 | **1.411** | 0.513 | 0.289 | 56.4% |
| 8 | 1.924 | 1.700 | **1.411** | 0.513 | 0.289 | 56.4% |
| 9 | 1.879 | **1.704** | 1.401 | 0.478 | 0.303 | **63.5**% |
| 10 | 1.879 | **1.704** | 1.401 | 0.478 | 0.303 | **63.5**% |
| 11 | 2.074 | 1.632 | 1.320 | 0.754 | 0.312 | 41.4% |
| 12 | 2.074 | 1.632 | 1.320 | 0.754 | 0.312 | 41.4% |
| 13 | 2.312 | 1.389 | 1.041 | 1.271 | **0.347** | 27.3% |
| 14 | 2.312 | 1.389 | 1.041 | 1.271 | **0.347** | 27.3% |
| 15 | 2.514 | 0.611 | 0.442 | 2.071 | 0.169 | 8.1% |
| 16 | 2.514 | 0.611 | 0.442 | 2.071 | 0.169 | 8.1% |

The same qualitative story as the Gaussian family holds, now with *four different* interior optima across four different columns: $V_{\mathrm{adaptive}}$ peaks at $m^\star = 9$, $V_{\mathrm{fixed}}$ peaks at $m = 7$, realized gain peaks at $m = 13$, and $\mathbb{E}[IG]$ is monotonically U-shaped (minimized at the middle and largest at the extremes $m = 1$ and $m = 16$, i.e., point-like and fully-isotropic lobes).

**Within-top-hat-family three-way comparison:**

| $\boldsymbol{c}$ | $\mathbb{E}[IG]$ | $V_{\mathrm{adaptive}}$ | $V_{\mathrm{fixed}}$ | realized | sat. |
|---|---|---|---|---|---|
| top-hat $m = 1$ (narrow extreme) | **2.081** | 0.619 | 0.499 | 0.121 | 5.8% |
| top-hat $m = 16$ (isotropic extreme) | 2.071 | 0.611 | 0.442 | 0.169 | 8.1% |
| top-hat $m = 9$ (in-family optimum) | 0.478 | **1.704** | **1.401** | **0.303** | **63.5**% |

The in-family optimum $m^\star = 9$ dominates both extremes on *every* actionable column—$V_{\mathrm{adaptive}}$, $V_{\mathrm{fixed}}$, realized gain, and saturation. The only column on which it is *not* best is $\mathbb{E}[IG]$, where both extremes beat it by a factor of four. An EVPI-only geometry selection would reject $m^\star = 9$ in favor of either $m = 1$ or $m = 16$, both of which are almost three times worse on $V_{\mathrm{adaptive}}$.

The top-hat-family $V^\star_{\mathrm{adaptive}} = 1.704$ nats slightly exceeds the Gaussian-family $V^\star_{\mathrm{adaptive}} = 0.865$ nats, confirming (numerically) that the top-hat family is Pareto-superior at this $K, N, p_{\mathrm{D}}, p_{\mathrm{FA}}$ for the in-family optimum. In an actual design setting, the realizable family is constrained by

the antenna hardware; the point of the numerics is that *given* a family, the joint DP recovers its optimum and EVPI does not.

# 15 Related work and originality check

Before deploying this case study in a paper, we surveyed the adjacent literatures to check whether the specific problem formulation—finite-POMDP ring target search with envelope-theorem-based antenna-geometry co-design and an EVPI pre-solver gate—has been published elsewhere. The summary below is based on a targeted web search across cognitive-radar, sensor-management, adaptive-compressed-sensing, sequential Bayesian DoE, end-to-end-optics-co-design, programmable-metasurface, sequential-hypothesis-testing, and information-directed-sampling literatures.

## 15.1 Closest prior work in each community

**Cognitive radar.**
The closest published work is Bouhou et al., *POMDP-Driven Cognitive Massive MIMO Radar* (arXiv:2410.17967, IEEE Trans. Radar Syst. 2025) and its follow-up *Joint Multi-Target Detection-Tracking via POMCP* (arXiv:2507.17506). These formulate cognitive MIMO radar as a POMDP but solve it approximately with Monte-Carlo tree planning (POMCP), adapt transmit-waveform and power, and *fix* the antenna aperture. Haykin's cognitive radar, Bell–Rangaswamy–Stinco, and Greco–Stinco–Gini's 2018 SPM survey all treat the aperture as given. No work in this community co-optimizes physical antenna element positions/excitations with an exactly-solved POMDP policy.

**Sensor management.**
The Hero–Cochran–Kreucher Foundations & Trends survey (2011) and the Hero–Castañón–Cochran–Kastella 2008 Springer volume cover POMDP sensor scheduling extensively (Kreucher–Kastella–Hero active sensing 2005; Castañón non-myopic classification). In every case the design variable is the scheduling policy over a fixed-menu sensor set; the physical hardware is never a co-design variable.

**Adaptive compressed sensing.**
Haupt–Castro–Nowak *Distilled Sensing* (IEEE Trans. Inf. Theory 2011) and Malloy–Nowak adapt *which rows* of an abstract sensing matrix to apply, without parameterizing the sensing matrix by physical array geometry and without POMDP DP.

**Sequential Bayesian DoE.**
Huan–Marzouk (arXiv:1604.08320, 2016) and Shen–Huan (arXiv:2110.15335, 2023) formulate sOED as a DP with continuous design spaces, solved approximately via ADP/policy-gradient RL. Alexanderian's 2021 *Acta Numerica* survey confirms the absence of exact Sondik-alpha-vector POMDP plus envelope-theorem physical-apparatus co-design in this community. The idea of EVPI / Expected Value of Sample Information *as a go/no-go gate* appears in clinical-trial VOI analysis (Flight–Julious–Brennan–Todd 2022) but not in radar/photonic sensor design.

**End-to-end optics co-design.**
Sitzmann et al. (SIGGRAPH 2018), Chang–Wetzstein 2019, Metzler–Heide–Wetzstein, Tseng–Majumdar, Kellman–Bostan–Waller (arXiv:1808.03571) jointly optimize static hardware with a *fixed, non-adaptive* reconstruction network or illumination sequence. No POMDP, no belief-state DP, no envelope-theorem policy/geometry split.

**Programmable metasurfaces / RIS.**
Compressive metasurface-aperture imaging (Yurduseven, Imani, Hunt, Smith) co-designs masks with reconstruction non-adaptively. RIS beamforming with deep RL uses Gaussian MIMO state models and abstract phase-shift layouts; hardware geometry is fixed.

**Sequential hypothesis testing.**
Chernoff (1959) and Naghshvar–Javidi *Active Sequential Hypothesis Testing* (Annals of Statistics 2013, arXiv:1203.4626) treat $K$-ary testing with adaptive experiment selection from a *fixed finite menu of actions*. The abstract structure matches our ring-search problem, but these papers do not parameterize the experiment's Fisher/Kullback matrix by continuous antenna geometry and do not identify closed forms like $\mathbb{E}[IG]_{\text{narrow}} = \frac{K-N}{K}\ln(K-N)$ or $\mathbb{E}[IG]_{\text{wide}} = \ln\big(K/(2N)\big)$.

**Information-directed sampling.**
Russo–Van Roy (JMLR 2018, arXiv:1403.5556) and recent work C-IDS on contextual POMDPs (arXiv:2602.03939) embed POMDPs in bilevel optimization, with abstract context variables and approximate policies—the closest *conceptual* match to our outer-context / inner-POMDP split—but not the envelope-theorem identity on the exact optimal value, and not the specific physical-geometry application.

### 15.2 What appears novel in the present framework

Individually well-established components: exact finite-POMDP DP via Sondik alpha-vectors (1971); adaptive radar beam scheduling with POMDPs; sequential Bayesian DoE as DP; envelope/Danskin theorem for bilevel optimization (Milgrom–Segal 2002); EVPI as a decision-theoretic quantity (classical Howard–Raiffa–Schlaifer); end-to-end differentiable hardware co-design (Wetzstein/Waller).

Combinations not found in the searched literature:

1. Treating antenna element positions and complex excitations as the outer design variable that *parameterizes the observation model of an exactly-solved finite POMDP*, with policy iteration alternating Sondik-exact Bellman and geometry gradient steps.
2. Using the envelope/Danskin identity on the optimal POMDP value so the geometry gradient reduces to the explicit partial at the fixed policy, avoiding implicit differentiation through the DP.
3. EVPI / ignorance gap as a pre-solver go/no-go gate with closed-form expectations $\mathbb{E}[IG]^{\text{narrow}}$ and $\mathbb{E}[IG]^{\text{wide}}$ under $D_K$-symmetric priors.
4. Orbit-symmetry ($D_K$ dihedral) decomposition of the per-cell $IG(x)$ distribution for the ring search, yielding the two-value (narrow) and five-value (wide) histograms we observe numerically.

### 15.3 Caveat

Web-search coverage of paywalled radar venues (IET, IEEE AES, NATO STO) and thesis repositories is uneven, and classified cognitive-radar literature is not searchable. If a direct prior art exists it is likely in an obscure venue or a very recent unindexed preprint. We found no direct match across the surveyed communities, and the specific closed-form EVPI predictions and orbit-dependent per-cell $IG(x)$ distributions appear not to have been reported previously.

## A Julia implementation

The following self-contained script produces all numerical values in this document. It is placed alongside this `.tex` file as `compute_ig.jl`.

```julia
const K = 16; const N = 4
const p_D = 0.9; const p_FA = 0.05
φ(z) = 2*pi * (z - 1) / K

function make_beam_coverage(m::Int)
    coverage = [falses(K) for _ in 1:K]
    half = div(m, 2)
    for a in 1:K, z in 1:K
        d = minimum((mod(z - a, K), mod(a - z, K)))
        coverage[a][z] = (m == 1 ? d == 0 :
                          (d <= half - 1) || (m % 2 == 1 && d == half))
    end
    coverage
end

p_detect(z, a, cov) = cov[a][z] ? p_D : p_FA

function posterior_entropy(s, y, cov)
    logb = zeros(K)
    for z in 1:K, k in 1:N
        pk = p_detect(z, s[k], cov)
        logb[z] += log(y[k] == 1 ? pk : 1 - pk)
    end
    m = maximum(logb); b = exp.(logb .- m); b ./= sum(b)
    -sum(b[z] > 0 ? b[z] * log(b[z]) : 0.0 for z in 1:K)
end

py_given_xs(y, x, s, cov) = prod(y[k] == 1 ? p_detect(x, s[k], cov) :
                                          1 - p_detect(x, s[k], cov) for k in 1:N)

function Φ_value(x, s, cov)
    lnK = log(K); val = 0.0
    for bits in 0:(2^N - 1)
        y = [((bits >> (k-1)) & 1) for k in 1:N]
        pyx = py_given_xs(y, x, s, cov)
        if pyx > 0
            val += pyx * (lnK - posterior_entropy(s, y, cov))
        end
    end
    val
end
```

The main loops (oracle and fixed enumerations) are straightforward four-nested loops over the pointings; see `compute_ig.jl` for the full listing.

# B Sanity check: $\mathbb{E}[IG]$ as a scalar

As a consistency test,

$$\mathbb{E}[IG]^{\text{narrow}} = V_{\text{oracle}}^{\text{narrow}} - V_{\text{fixed}}^{\text{narrow}} = 2.579634 - 0.498683 = 2.080951$$

matches the independently-computed $\mathbb{E}_x[IG(x)] = 2.080950$ to five decimal places (the sixth is floating-point), confirming the arithmetic. Similarly in the wide regime:

$$1.923462 - 1.410526 = 0.512936 \;\approx\; 0.512935.$$